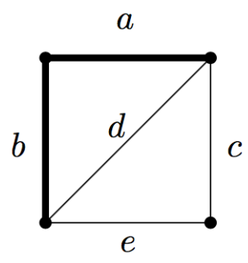
Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 60%. We keep your highest score.

Next item →

1. Consider the graph on four vertices below. The two fat edges a, b have weight 1 each, the remaining three edges c, d, e have weight 2 each. Which of the following edge sets are *good*, i.e., are contained in some minimum spanning tree? 1 / 1 point

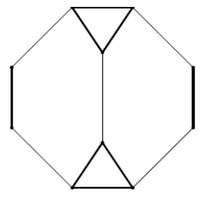


Nice work

2. Let G be a connected graph on $n \geq 1$ vertices with a cost function on its edges. Which of the following statements are true? 1 / 1 point

Nice work

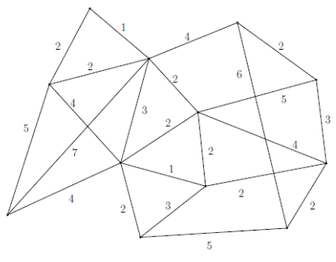
3. In the graph below, fat edges have weight 1 and normal edges have weight 2. 1 / 1 point



How many minimum spanning trees does it have?

Nice work

4. Let $G = (V, E)$ be a graph and $c : E \to \mathbb{R}^+$ be an edge cost function. We call an edge $e \in E$ (1) *necessary* if e is contained in *every* minimum spanning tree; (2) *useless* if no minimum spanning tree contains e; (3) *optional* else. How many edges in the graph below are necessary? 1 / 1 point



Nice work

5. How many edges of the above graph are optional? 1 / 1 point

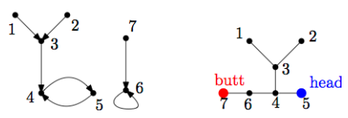
Nice work

6. How many edges of the above graph are useless? 1 / 1 point

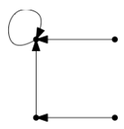
Nice work

7. The centerpiece of our proof that K_n has n^{n-2} spanning trees was a correspondence between functions $f : [n] \to [n]$ and *vertebrates*, that is, spanning trees with a *head* h and a *butt* b. 1 / 1 point

x	1	2	3	4	5	6	7
$f(x)$	3	3	4	5	4	6	6



Let us define the *core* of a function as the set of all elements $i \in [n]$ "lying on a cycle". For example, $\text{core}(f) = \{4, 5, 6\}$ in the above picture. More formally we can define core as the image of f^n. It is easy to see that every function has a non-empty core. The above function has a core of size 3. The function below has a core of size 1:



Let $c(n)$ denote the number of functions $f : [n] \to [n]$ with a core of size 1. The peer review asks you to come up with an explicit formula for this.

What is $c(3)$?

Nice work

8. What is $c(4)$? 1 / 1 point

Nice work

9. What is $c(29)$? 1 / 1 point

Nice work